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Hydrophobic ionic liquid enabled polyiodide confined transport in a cathode, realizing high areal capacity, stable zinc–iodine batteries

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Aqueous zinc–iodine batteries are promising energy storage candidates due to their high safety and moderate cost. A high areal-capacity iodine cathode is the key to achieving practical batteries towards commercialization. However, high iodine loading exacerbates polyiodide shuttling and reduces the electrical conductivity of the electrodes. The traditional solution to these problems using a porous carbon matrix has a limited effect towards shuttling and leads to low volumetric energy density. In this study, we proposed a hydrophobic liquid ionic conductive agent to mediate confined iodine transport in a thick electrode, realizing a highly stable zinc–iodine battery with ultra-high iodine mass loading. A zinc–iodine battery with an ionic liquid conductive agent demonstrates an increase to 130% of the volumetric capacity compared with traditional porous carbon cathodes, reaching a 7-fold increase in cycle life (3500 cycles) at 10 mg cm^{−2} iodine loading and a 40-fold increase (2000 cycles) at 40 mg cm^{−2}, with an initial areal capacity of 6.30 mAh cm^{−2} and a capacity decay of 0.00495% per cycle. An ultra-high loading of 60 mg cm^{−2} with an areal capacity of 11.55 mAh cm^{−2} (193 mAh g^{−1}) can also be achieved. Consequently, the Ah-level Zn–I₂ pouch cell assembled with a highly loaded thick electrode demonstrates excellent cycling stability, substantiating its potential for practical application.

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Broader context

Zinc batteries based on iodine cathodes have emerged as one of the most promising candidates for various energy storage devices. To be comparable to lead-acid batteries, it is imperative for a practical zinc–ion battery cathode to achieve a high areal capacity, which requires a thick electrode with high mass loading of iodine. However, increasing the iodine cathode thickness and mass loading brings serious problems to iodine utilization, rate capability, and cycle life. These problems can be assigned to the fast decrease of electrode conductivity and the aggravation of the shuttle effect with increasing areal iodine loading. The conventional strategy of only using conductive carbon with low tapping density, in highly loaded electrodes, will only lead to an increase in the electrode thickness. In this study, we proposed a hydrophobic liquid ionic conductive agent (LICA) to mediate confined iodine transport in a thick electrode, for realizing a highly stable zinc–iodine battery with high areal capacity. This strategy can optimize ionic conduction while suppressing shuttle effects, thereby ensuring high stability at ultra-high iodine loadings. This study offers novel design concepts for the fabrication of highly loaded thick electrodes and provides evidence supporting the viability of Zn–I₂ batteries in real-world applications.

Introduction

Zinc batteries based on iodine cathodes have emerged as one of the most promising candidates for various energy storage devices.^{1–4} The iodine cathode has moderate price (\$6500/ton),

high specific capacity (211 mAh g^{−1}) and low toxicity, which is promising for commercialization.^{5–9} One of the most unavoidable challenges in a zinc–ion battery cathode is achieving a high areal capacity above 5 mAh cm^{−2} to match the ultra-high areal capacity of the zinc anode (11.7 mAh cm^{−2} at 20 μm thickness) and a high volumetric capacity of above 63.6 mAh cm^{−3} to be comparable with lead-acid batteries, which require both a thick electrode and a high mass loading of iodine.^{10–14}

However, merely increasing iodine cathode thickness and mass loading brings serious problems in iodine utilization, rate capability, and cycle life.^{15–17} These problems can be attributed to the fast decrease of electrode conductivity and the

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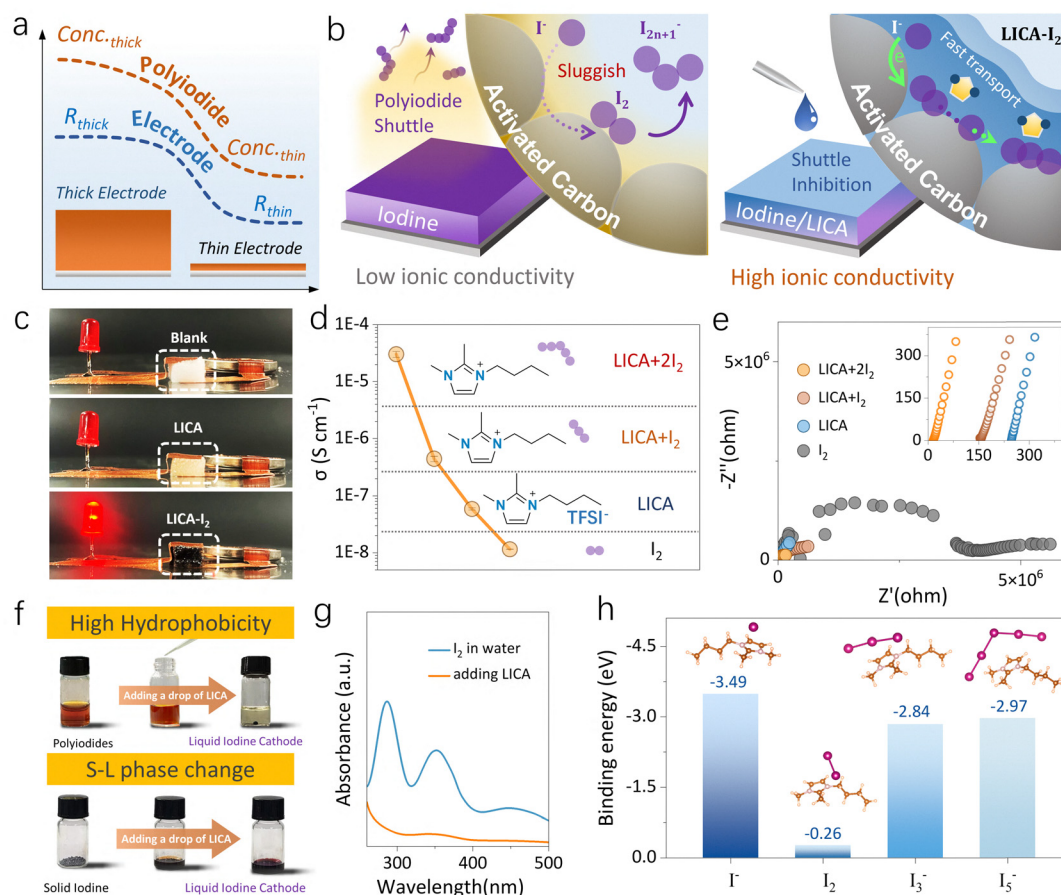


Fig. 1 (a) The comparison between the thick electrode and the thin electrode. (b) Schematic illustration of the iodine deposition behavior in the traditional iodine cathode and novel LICA-I₂ cathode. (c) A simple demonstration of the electrical conductivity of LICA combined with iodine: the LED bulb lights up immediately after connection to a copper strip. (d) The electrical conductivity of pure iodine, LICA, and the complex with different iodine content, measured by using a potentiostatic test at 25 °C. (e) Electrochemical impedance spectroscopy (EIS) results of pure iodine, LICA, and the complexes. (f) Demonstration of hydrophobicity and solid-liquid phase change of LICA combined with iodine. (g) UV-vis spectroscopy of I₂ aqueous solution before and after adding LICA. (h) The optimized adsorption configurations and corresponding binding energies of I⁻, I₂, I₃⁻, and I₅⁻ near the imidazole cation.

aggravation of shuttle effect with increasing areal iodine loading, as shown in Fig. 1a.^{18–20} Iodine is an electrically/ionically insulating element ($\sim 10^{-7}$ S m⁻¹).²¹ Therefore, iodine in poor electric contact with conductive carbon is difficult to undergo reduction, which leads to low utilization of iodine and eventually reduces the overall energy density. The shuttle effect is another problem that occurs in a thick iodine electrode. According to Fick's law, the diffusion flux is determined by the concentration gradient.²² High iodine areal loading leads to higher concentration of polyiodide, and therefore the polyiodide shuttle would be more likely to take place. A more severe shuttle effect of a high-loading electrode leads to serious self-discharge as well as sudden failure of the battery with a shortened operating life.

Conductive agents with porous structures such as activated carbon and carbon black are classic solutions to enhance electrical conductivity and to reduce the shuttle effect.^{23–25} The high electron transport in these carbon-based materials remarkably facilitates the redox charge transfer, while the micropores effectively adsorb iodine/polyiodide molecules to

reduce the shuttle effect. The porous conductive agent is highly efficient in improving the capacity as well as cycle stability in thin electrode conditions. However, when it comes to high loading electrodes, the introduction of a large amount of low tapping density conductive carbon (~ 0.25 g cm⁻³) results in mere increase of electrode thickness.¹¹ Therefore, more effective strategies should be developed to enhance the overall conductivity in the case of thick electrodes.

Ionic conduction is another crucial issue in contributing overall conductivity other than electronic conduction, especially in iodine cathodes. The ionic transport of iodine species comprises a unique Grotthuss-like process with the iodine exchange reaction: I⁻...I₃⁻...I⁻. Therefore, enhancing the kinetics of the iodine exchange reaction can effectively facilitate the ionic transport in iodine cathodes. Ionic liquids (IL) usually have high density (~ 1.42 g cm⁻³) and have been applied to offer ionic conductivity in various applications.^{26–31} Specifically, it was reported that the ionic conductivity of both iodine and IL could be enhanced by formation of polyiodide IL (PIIL)

due to the soft acid–base interaction between quaternary ammonium cations and polyiodide anions.^{32–34} Besides, the PIIL is water-insoluble due to the hydrophobicity of organic cations. Therefore, highly conductive and shuttle-free thick electrodes can be achieved when an IL is introduced as a liquid ionic conductive agent (LICA) (Fig. 1b).

Herein, as a proof of concept, we introduce a hydrophobic ionic liquid 1-butyl-2,3-dimethylimidazolium bis((trifluoromethyl)sulfonyl)imide (BMMImTFSI) as the cathode LICA, which forms a highly reactive liquid network (HRLN) that connects the pores and crevices throughout the thick electrode. The LICA spontaneously transforms the insulating solid iodine into highly conductive liquid state polyiodides *via* strong interaction of Lewis acidic iodine and Lewis basic pseudo-halogen TFSI[−] anions, which remarkably enhance the thick electrode conductivity.³⁵ Moreover, the large BMMIm⁺ cation binds the polyiodides generated during charge/discharge cycles, and hence significantly reduces the shuttle effect and prolongs the cycle life of the thick electrode. Besides, the hydrophobic LICA is free of leakage in the electrolyte during the entire charging and discharging process. Benefiting from these advantages of these unique properties of LICA, the cathode with confined iodine transport realized an ultra-high loading of 60 mg cm^{−2} with an ultra-high areal capacity of 11.55 mAh cm^{−2}. The lifespan of the battery has also been greatly improved, reaching a 7-fold increase in cycle life (3500 cycles) at 10 mg cm^{−2} iodine loading and a 40-fold increase (2000 cycles) at 40 mg cm^{−2}, with an initial areal capacity of 6.30 mAh cm^{−2} and a capacity decay of 0.00495% per cycle. As a demonstration for practical application, an Ah-level Zn–I₂ pouch cell with an LICA assisted cathode with high mass loading also showed stable cycle life.

Results and discussion

Interactions between liquid conductive agent and iodine species

First, a circuit was designed and constructed using copper tape, with a connection point in the middle that was filled with a piece of sponge, shown in Fig. 1c. The circuit was in a broken state at the outset, and the bulb could not be illuminated. When LICA was added to the sponge, the bulb remained unlit, as the conductivity of LICA itself was insufficient. However, after the addition of the LICA–I₂ complex, the bulb was successfully illuminated, thereby demonstrating its remarkable conductivity. In order to further investigate the effect of the introduction of a liquid conductive agent on the ionic conduction, the electrical conductivity of the LICA and its complexes were measured³⁶ (Fig. 1d and Fig. S1). The electrical conductivity of pure iodine is 1.2×10^{-8} S cm^{−2} at 25 °C. The addition of the LICA has been shown to result in a significant difference in conductivity, exhibiting 1.5 orders of magnitude increase compared with pure iodine. Furthermore, as the amount of iodine added increases, the conductivity increases by 3.5 orders of magnitude (3.0×10^{-5} S cm^{−2} at 25 °C), which can be explained by the appearance of higher polyiodides giving rise to

increased bond exchange.^{32,33} The charge-transfer impedance (R_{ct}) was obtained *via* electrochemical impedance spectroscopy (EIS), shown in Fig. 1e. The results show that R_{ct} of LICA–iodine complexes are 4–5 orders of magnitude lower than that of pure iodine, suggesting higher ionic mobility at room temperature. What's more, the addition of iodine also reduces R_{ct} of a pure liquid conductive agent, due to the acceleration of ion conduction by the Grotthuss bond-exchange process of polyhalides, which has been widely applied in dye-sensitized solar cells.^{37–39} It is noteworthy that Grotthuss-type electron-exchange conduction or polyiodide bond-exchange conduction can occur, in such a manner that charge transport takes place *via* electron exchange or bond exchange between closely packed I[−] and I₃[−] ions, rather than *via* long-range effective mass transfer.⁴⁰ During this process, electron hopping and polyiodide bond exchange are coupled.⁴¹ The imidazole cations in the LICA stabilize the structure of the polyiodide chain/network through π – π stacking and charge transfer, providing a low-energy, continuous pathway for the valence-changing ‘jumping’ conduction.⁴²

The ionic liquid conductive agent has excellent binding and extractive properties for iodine. Initially, the LICA itself is colorless and transparent. Following the addition of water containing litmus to the bottle containing the LICA, a clear delamination occurs, as is demonstrated in Fig. S2. This observation serves to illustrate the hydrophobicity of LICA itself. The addition of a single drop of colorless LICA to I₃[−] solution results in the immediate generation of a multitude of minute droplets, shown in Fig. 1f. Following a period of 12 hours, the aqueous solution is observed to have a near-transparent appearance, with the presence of a dark brown hydrophobic droplet, which is identified as the liquid conductive agent after absorbing I₃[−]. It is proven that the ionic liquid conductive agent firmly adsorbs both I₂ and I₃[−], thereby reducing the loss of active species and obtaining a higher efficiency. When we mix it directly with iodine to form an LICA–iodine complex, a solid–liquid phase change was seen. The Raman spectra of LICA–iodine complex is shown in Fig. S3, showing a difference near 206 cm^{−1}, which can be assigned to iodine with certain interaction with the liquid conductive agent.⁴³ The application of these properties to the battery results in the maintenance of a stable state during the charging and discharging processes in the vicinity of the cathode. Similarly, when adding the LICA to an aqueous solution of iodine, the yellowish color in the aqueous solution fades to transparency. In contrast, the liquid turns yellow and stays stratified with the aqueous phase, as shown in Fig. S4. Using UV-vis spectroscopy (UV-vis) to test the composition of the water phase (Fig. 1g), the characteristic absorption peaks of I₃[−] at 290 and 350 nm obviously vanish.⁴⁴ The I₂ peak around 450 nm also disappears significantly.²⁵ In addition, the adsorption energies between imidazole cation of LICA and various iodine species can be measured using density functional theory (DFT) calculations. The optimized adsorption configurations and corresponding binding energies of the four iodine species (I[−], I₂, I₃[−], and I₅[−]) near the imidazole cation are shown in Fig. 1h. The binding energies between

imidazole cations containing electron-withdrawing groups and iodine species are all negative, suggesting that all the adsorption behaviors are thermodynamically favored. A comparison of binding energies with water molecules (Fig. S5) indicates the stronger binding affinity between iodine species and the imidazole cation than with H_2O . This further demonstrates that LICA can adsorb iodine species and form stable complexes, effectively inhibiting the binding of iodine with water, thereby stabilizing free polyiodides. This mechanism significantly alleviates the polyiodide shuttle effect in Zn-I_2 batteries.

Ionic conduction along the perpendicular direction of thick electrodes

The construction of thick electrodes can result in a reduction of inactive materials at the battery device level, an improvement in energy density, and a reduction in battery cost. However, the inherent limitations of conventional thick electrodes, including slow ion and electron transport, result in rate and cycling performance. Here, we initially concentrate on the ion transport in the perpendicular direction for thick electrodes.

The Raman spectra of the electrode surface after the completion of charging process in a low concentration electrolyte

are presented in Fig. S6. The spectrum of the LICA/AC electrode displays the characteristic peaks of I_3^- , I_5^- , I_2 , and the functional groups of TFSI^- . A series of curves are obtained by line scanning of the cross-section of the electrode (Fig. 2a and b). It is observed that for the AC electrode, the peaks corresponding to iodine species can only be detected near the surface, whereas for the LICA/AC electrode, signals belonging to iodine species can be detected at almost all depths of the electrode. This is due to the slow diffusion and transport of iodine species ions in the AC electrode, resulting in the primary formation and accumulation of charging products on the electrode surface. In contrast, the liquid ionic conductive agent in the LICA/AC electrode accelerates ion transport within the electrode, significantly enhancing ion transport and diffusion efficiency in the vertical direction. This allows ions to migrate further away from the surface to undergo iodine conversion reactions, while the generated charging product ions can also diffuse deeper into the electrode, thereby achieving more uniform and thorough dissolution–deposition reactions within the LICA/AC thick electrode. Furthermore, the elemental distribution of LICA/AC following the charging process is analyzed using Time of Flight Secondary Ion Mass Spectrometry (ToF-SIMS)

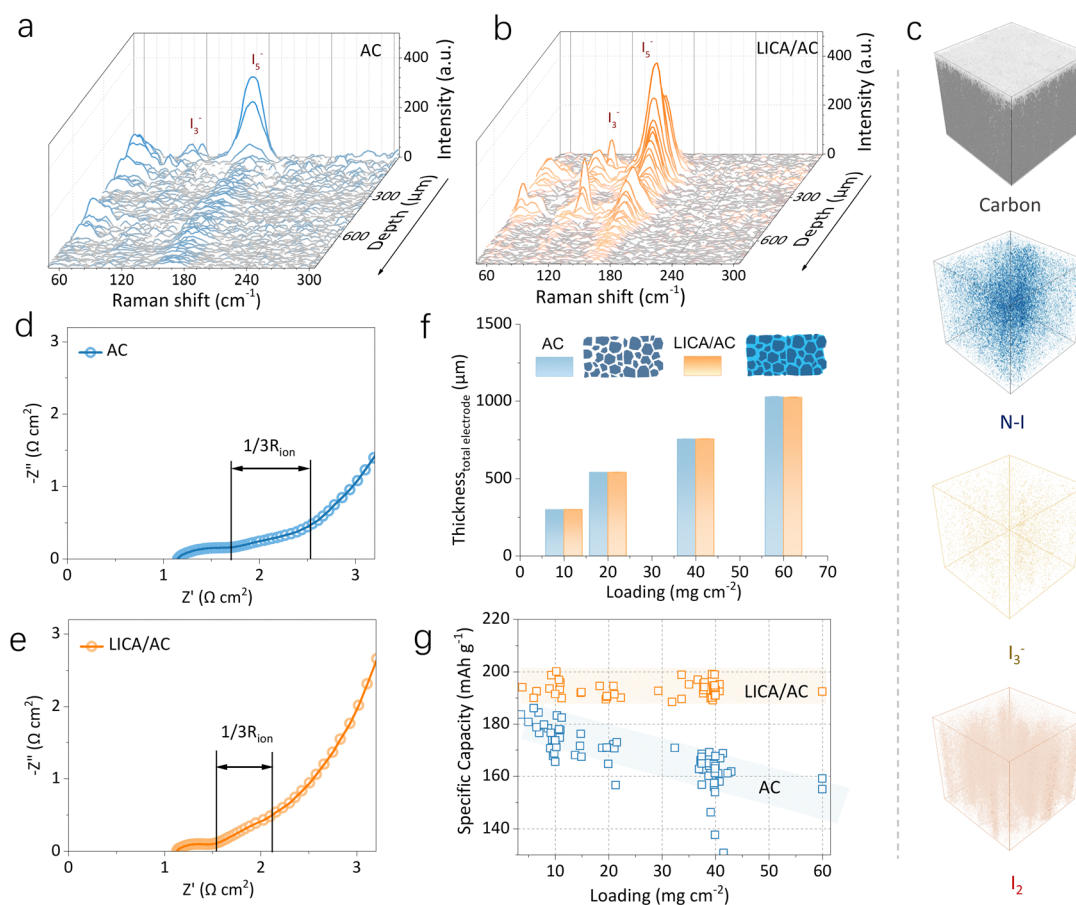


Fig. 2 (a) Line scanning Raman spectra of the cross-sections of AC thick electrode and (b) LICA/AC thick electrode after charging. (c) 3D render overlay for elements of time-of-flight secondary ion mass spectrometry (ToF-SIMS) after iodine deposition on LICA/AC. (d) Nyquist plots of symmetric cells using two identical AC electrodes at 0% SOC, and (e) two identical LICA/AC electrodes at 0% SOC. (f) Thicknesses of different electrodes (including current collectors) with different iodine loadings. (g) Specific capacities of LICA/AC at different iodine loadings.

(Fig. 2c and Fig. S7). The distribution of charging products can be referenced to I_3^- , I_5^- , and I_2 , all of which are observed to be homogeneously distributed. Thanks to the uniform distribution of iodine species, it can fully interact with the porous carbon, which serves as both host and conductive agent, suppressing the shuttle effect while maintaining good conductive contact. This facilitates the discharge process, leading to enhanced coulombic efficiency and increased discharge capacity. Furthermore, N-I signals are detected, which serves to corroborate the assertion that the complexation of iodine species with liquid conductive agent is primarily contingent upon the interaction between N and I.

The characteristics of the pore structure of the thick electrode are elucidated through small-angle X-ray scattering (SAXS). The scattering intensity of SAXS is proportional to the number of scattered photons; thus, the higher the electron density contrast between substances, the higher the intensity.^{45,46} The LICA/AC electrode results in a notable reduction in the scattering intensity in comparison to the AC electrode, as illustrated in Fig. S8. This indicates that the liquid conductive agent fills the pore structure, which will reduce the electron density difference between substances.

The electrochemical process in an electrode structure can be divided into several stages: (1) ion transport in bulk electrolyte; (2) the electric double layer at the interface between the electrolyte and the electrode; (3) ion diffusion in the pores of electrode (corresponding to R_{ion}); and (4) charge transfer reaction for active species.^{47,48} In batteries with thin electrodes, the charge-transfer impedance (R_{ct}) represents the rate-determining step in the kinetic behavior within the battery. Conversely, in thick electrodes, the ion transport impedance (R_{ion}) gradually becomes the rate-dependent step as the thickness of the electrodes increases.⁴⁹ The EIS measurement of a symmetric cell (EIS-SC), comprising identical electrodes, enables the decoupling of electrode internal resistances, thereby offering significant advantages in comparison to the conventional EIS test for half/full cells. This can be explained by the theory of transmission line model (TLM).^{49,50} Fig. 2d and e illustrate the Nyquist diagrams for AC and LICA/AC electrodes with identical loadings at 0% state of charge (SOC). The curves comprise a distorted semicircle, a straight line with a slope close to 45° from the actual axis, and an inclined line with a markedly larger slope. It should be noted that the curve in question does not represent R_{ct} , as there is no Faraday process occurring. In the figure, the diagonal line close to 45° theoretically reflects the resistance to ion transport inside the porous electrode and can be represented by R_{ion} . The values of R_{ion} of LICA/AC are all less than those of AC, indicating fast ion transport within electrodes. Fig. 2f illustrates that, under varying iodine loadings, the overall thicknesses of LICA/AC and AC electrodes remain consistent, indicating that the incorporation of ionic liquids has no discernible impact on electrode thickness, thus contributing to the enhancement of volumetric capacity. Furthermore, Fig. 2g illustrates that, across multiple tests, cells with LICA/AC electrodes can sustain a relatively consistent specific capacity despite varying active substance loadings.

The increase in electrode thickness does not diminish the utilization of the active substance, which contributes to the achievement of higher areal capacity at high iodine loading. Conversely, the specific capacity of electrodes only using porous carbon as both conductive agent and adsorbent, exhibits a substantial decline with increasing active substance loading and electrode thickness.

Optimization of the electrochemical behavior of iodine species

Based on aforementioned properties, an in-depth examination of the impact of LICA on the electrochemical process is investigated. First, we use the microelectrode to measure cyclic voltammetry (CV) curves for analytical purposes, shown in Fig. S9. For pure NaI solution, the oxidation process is accompanied by the rapid formation of solid I_2 on the electrode surface, due to the low solubility of I_2 in aqueous solution. Following an initial increase, the oxidation current begins to decrease after reaching a value of 26 nA. This is attributed to the poor conductivity of I_2 , which limits the transfer of charge on the electrode surface.⁵¹ Ultimately, a steady state is attained through the conversion of I_2 into I_3^- , resulting in the gradual stabilization of the oxidation current. It is also difficult to observe the reduction peak in pure NaI, which is due to the formation of soluble I_3^- , which dissolves into the bulk solution and cannot be detected. Fig. S9 provides a visual representation of the aforementioned phenomenon. After adding LICA, an obvious reduction peak can be observed, with a current of 179.8 nA. In this instance, the liquid droplet of polyiodized ionic liquids, formed by oxidation, exhibits a tendency to adhere to the surface of the microelectrode, and its high conductivity makes it easy to exchange ions and electrons with the electrolyte and electrode. Furthermore, the hydrophobic nature of the ionic liquids ensures the stability of the reverse reduction process.

We additionally investigate the effect of LICA during iodine deposition. According to the galvanostatic charge/discharge curves in Fig. S10, the cell with liquid conductive agent has a higher areal capacity (2.47 mAh cm^{-2}), compared with the cell without LICA (1.5 mAh cm^{-2}). As illustrated in Fig. 3a, the enlargement of the beginning of the discharge curve reveals that the battery with the common carbon cathode exhibits a distinct overpotential of 80 mV. This is a consequence of the phase transition, indicating that the solid nucleation is situated on the current collector. In contrast, the battery with LICA exhibits no overpotential due to its liquid properties. Furthermore, the introduction of ionic liquids as conductive agent has resulted in a notable reduction in battery polarization, from 50.1 to 23.7 mV, indicating an advantage in electrochemical kinetics (Fig. 3b and Fig. S10). In terms of the active substance utilization and capacity retention, cells with LICA demonstrate considerable advantages, as shown in Fig. 3b and Fig. S11. Here the active substances are all derived from the electrolyte, so the discharge capacity can be converted to the utilization rate of iodide. In a more intuitive approach, we add a drop of liquid conductive agent onto the carbon cathode during the cycle process and continue the cycle, shown in Fig. 3c. After iodine

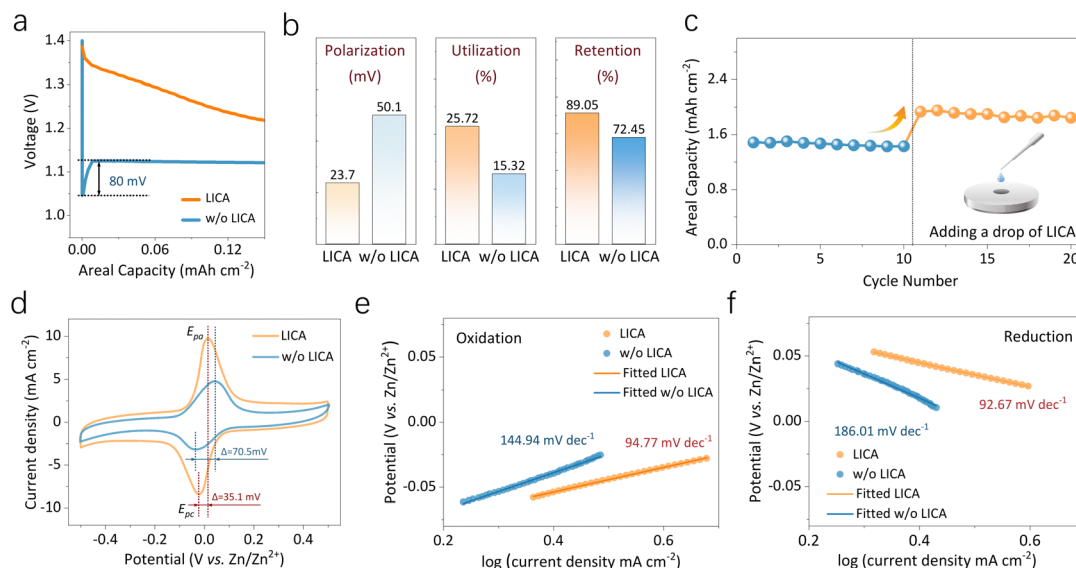


Fig. 3 (a) The enlarged voltage–capacity curves of batteries with and without LICA. (b) Comparison of polarization, utilization and capacity retention. (c) The change of areal capacity after adding a drop of LICA onto the cathode during cycling. (d)–(f) CV curves of symmetric batteries and corresponding Tafel plots from CV curves.

species combine with LICA is dropped on the electrode, the high conductivity of complexes significantly improves the efficiency of ion transport, the kinetics of redox reactions, and reversibility. This enables the iodine redox reaction to proceed more thoroughly. More active iodine species participate in effective charge transfer, ultimately achieving a higher discharge capacity. Furthermore, batteries with LICA outperform those without LICA in terms of coulombic efficiency, rate performance and self-discharge performance, showing the capability of inhibiting shuttle effects (Fig. S12–S14). The average coulombic efficiency of cell without LICA during 200 cycles is only 90.08%, while that of LICA reaches 96.23%, proving that the shuttle effect can be suppressed well. The kinetics of the iodine conversion reaction can be reflected by the rate performance. The cell with LICA shows a great improvement in the areal capacity and coulombic efficiency at current densities ranging from 1 to 8 mA cm⁻², and it achieves an areal capacity of 2.4 mAh cm⁻² at 1 mA cm⁻², which is at the higher level than those reported in the literature. Increasing the current density to 8 mA cm⁻² and bring it back to 0.5 mA cm⁻², the capacity of the cell returns to 2.4 mAh cm⁻², demonstrating the excellent electrochemical reversibility on the current collector with LICA. Following several cycles, the battery is fully charged and subsequently discharged after resting for a period of time. During resting, I₂ decomposition and polyiodide shuttling result in a reduction in the discharge capacity and energy efficiency. The battery with LICA can maintain higher energy efficiency after resting for 1 to 16 h, demonstrating superior storage performance. It is worth noting that similar soft acidic pyrrolidinium-based hydrophobic ionic liquids can also be introduced into the cathode as liquid conductive agents, similarly improving electrochemical performance (Fig. S15).

CV curves of symmetric cells are measured to further study the kinetics of the iodine redox (Fig. 3d). Both of them show excellent electrochemical reversibility. At an identical scan rate, the potential difference $|E_{pa} - E_{pc}|$ between the oxidation and reduction peaks of the common carbon cathode is double that of the cathode containing the LICA. This finding indicates that the cathode containing the LICA exhibits a weaker redox polarization. The results are consistent with those conducted above. The Tafel slope (η) can be calculated by plotting the potential against the logarithm of the current density, shown in Fig. 3e and f. The calculated η for cathode with LICA during oxidation (94.77 mV) and reduction (92.67 mV) are lower than those of common carbon cathode (oxidation, 144.94 mV; reduction, 186.01 mV), suggesting an accelerated redox reaction.

Cathode reaction with LICA

In order to monitor the evolution of the microstructure during the cycle and explore the intrinsic redox chemistry, a series of *in situ* and *ex situ* spectroscopy and electrochemical impedance tests are conducted. Initially, X-ray photoelectron spectroscopy (XPS) spectra are conducted on the cathode with LICA. As illustrated in Fig. 4a, the spectra exhibit a pronounced peak associated with I 3d following the charging process to 1.4 V. The high-resolution N 1s spectrum can be deconvoluted into two peaks at approximately 399.46 and 401.87 eV (Fig. 4b). However, an additional peak can be observed at 400.67 eV after charging the cell to 1.4 V, while the original two peaks of N 1s are shifted to high binding energies (399.60 and 401.89 eV). These findings can be attributed to the interaction of the generated iodine species with N. Furthermore, the high-resolution spectra of I 3d of cathode with LICA at 1.4 V display two strong peaks, corresponding to I 3d_{3/2} and I 3d_{5/2}, respectively (Fig. S16). The deconvolution of each peak into two

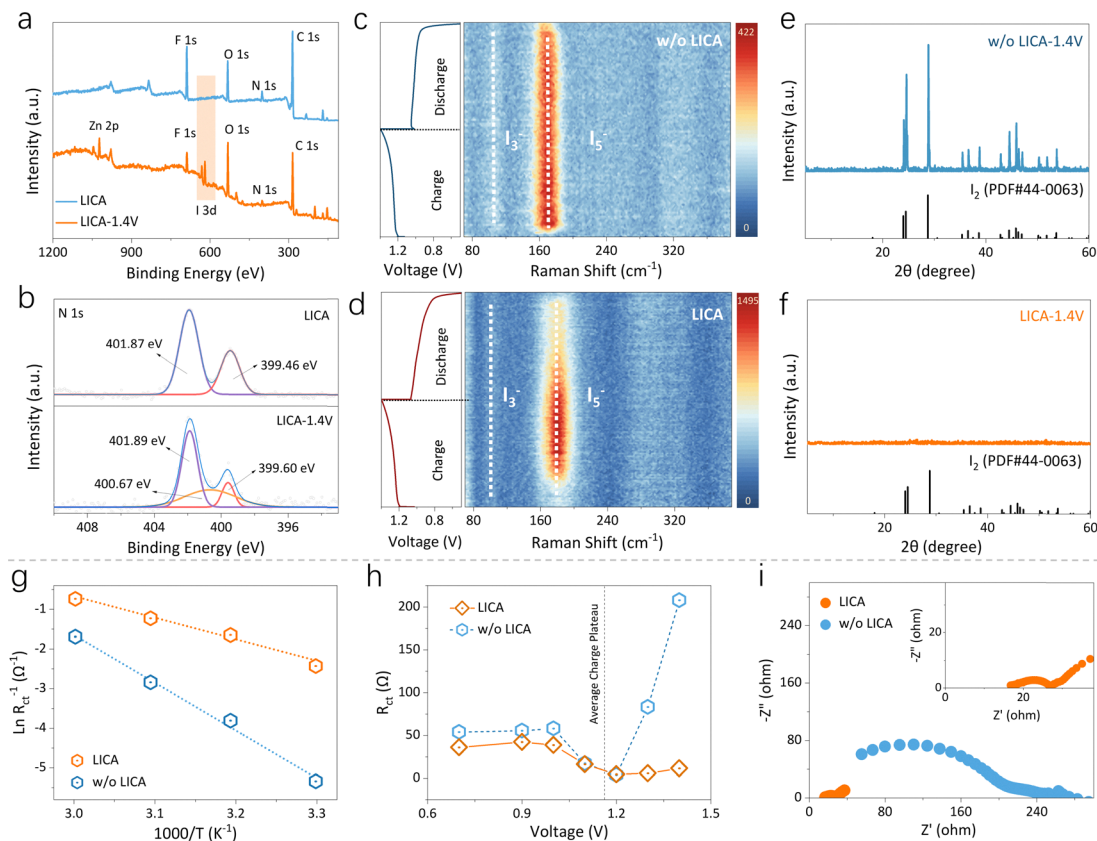


Fig. 4 (a) X-ray photoelectron spectroscopy (XPS) spectra, and (b) detailed N 1s XPS spectra of LICA-modified cathode at open-circuit voltage and at 1.4 V. (c) and (d) *In situ* Raman spectroscopy and the corresponding charge/discharge curves of cathode with and without LICA. (e) and (f) XRD patterns of cathode with and without LICA at 1.4 V. (g) Arrhenius curves fitted from the series of R_{ct} at different temperatures, which given the activation energy. (h) Impedance changes during the charging process. (i) EIS plots at 1.4 V. The inset is partially magnified EIS plots.

implies the presence of a subtle difference in the iodine valence, which implies the presence of a mixture of different iodine species.^{52,53}

To obtain insight into the charging and discharging mechanisms occurring during the electrochemical process, we employ *in situ* Raman spectroscopy to observe the changes in the polyiodide signals. In the case of common carbon cathode, the dominant signal is I_5^- (approximately 170 cm^{-1}), with a minor contribution from I_3^- (approximately 105 cm^{-1}). In the case of LICA, the I_5^- peak can be observed to undergo a red shift, reaching a value of 179 cm^{-1} . The intensity of the peak corresponds to the amount of polyiodide in the vicinity of the cathode. The I_5^- peak intensity of cell with LICA is considerably higher, which implies that a greater quantity of polyiodides is produced at the LICA-modified cathode. This finding is in accordance with the results of the areal capacity presented in Fig. S10. The intensity of the I_5^- Raman signal from the beginning of charging to the end of discharge in Fig. 4c remains robust throughout, indicating the presence of soluble polyiodide formation throughout the process, which would intensify its movement towards the anode, driven by the electric field, and react at the anode, resulting in the loss of the active substance, *i.e.* the “shuttle effect”. This will significantly influence the cycle life of cells. However, for the LICA-modified cathode in Fig. 4d, the

intensity of the I_5^- signal increases significantly upon reaching the plateau of the charging process, and then decreases abruptly at the onset of the discharge process. The accelerated kinetics facilitate the prompt conversion of polyiodides to the discharge product I^- during the discharge process, thereby circumventing the shuttle effect of polyiodide. X-ray diffraction (XRD) measurements are employed to assess the chemical composition of cathode at 1.4 V (Fig. 4e and f). The XRD spectrum of common carbon cathode exhibits multiple diffraction peaks, indicative of the presence of solid iodine as oxidation product. However, the spectrum of LICA-modified cathode exhibits no diffraction peaks. The combined analysis of the Raman and XRD results indicates that, the oxidation product of the cathode containing LICA changes to a composite, non-solid polyiodide product, which can be attributed to the binding of the liquid conductive agent to iodine.

The origin of the improved electrochemical performance can be studied by evaluating the charge transfer resistance (R_{ct}) through the EIS spectrum of cells with and without LICA. The charge transfer process represents a pivotal step in the transportation of ions and electrons to the active center, which ultimately determines the kinetics of the iodine redox reactions. By measuring the R_{ct} at different temperatures, it is possible to calculate the activation energy, which represents the reaction energy barrier, using the Arrhenius equation (see

Fig. 4g and Fig. S17).^{54,55} From the slope, it can be inferred that the electrode reaction after introducing the ionic liquids as a conductive agent exhibits a reduced activation energy, thereby indicating enhanced kinetics. Furthermore, we test the R_{ct} throughout the charging process, shown in Fig. 4h and Fig. S18. In the beginning, R_{ct} of the cell with LICA is only slightly lower than that of the cell without LICA; however, as the reaction proceeds, the difference becomes increasingly pronounced. The process of iodine deposition commences in proximity to the charge potential, as the iodine produced through oxidation can combine with I^- contained within the electrolyte, thereby forming a polyiodide that is soluble in the aqueous phase of the electrolyte. This results in a transient decline in the internal resistance within the cell.⁵⁶ As the charging process continues, a significant quantity of oxidation products will be deposited at the cathode, leading to an increase in the battery's internal resistance. The EIS curve of the battery during the charging process to 1.4 V is illustrated in Fig. 4i. It can be observed that the impedance of the battery without LICA at 1.4 V is 20 times higher than that of the battery with LICA. In conjunction with the aforementioned findings, a considerable quantity of solid iodine products is observed on the collector when the battery without LICA is fully charged. Low conductivity of solid iodine leads to ultra-high R_{ct} . In contrast, for the battery with LICA, the charging products at 100% SoC are primarily polyhalogenated ionic liquids, which can be confirmed by *in situ* Raman results in Fig. 4d. The charging products here are in a liquid state with high conductivity and are conducive to the redox reaction.

Construction of a thick electrode with LICA

In light of the preceding discussion, we prepare high-loading thick cathodes (LICA/AC) based on the ionic liquid conductive agent, successfully constructing the highly reactive liquid network (HRLN) which is capable of connecting the pores and crevices within the electrode. A low-cost, low-density, low-concentration electrolyte is employed for the subsequent tests. Scanning electron microscopy (SEM) images and corresponding EDS mappings of LICA/AC thick electrode are shown in Fig. S19–S21. SEM images show that no obvious cracks or delamination are observed on the electrode surface, indicating good structural integrity. The cross-sectional SEM images reveal the thickness and compactness of the thick electrode. Further elemental mapping reveals that various elements are uniformly distributed throughout the electrode. Since element N is solely derived from the LICA, its uniform distribution in the cross-section confirms that the ionic liquid has penetrated the entire thickness of the electrode, forming a continuous ionic conduction network. X-ray computer tomographic (XCT) images offer insight into the electrode structure and visualize the different components of the electrode.^{57,58} The distribution of voids within the electrode is clearly discernible from the 3D structure image and sliced images (Fig. S22). The use of different colors to delineate the various areas makes it evident that the electrode comprises the carbon matrix, liquid component and iodine, which are evenly distributed, as shown in

Fig. 5a. The AC and LICA/AC electrodes are clamped with stainless steel sheets and the conductivity of the electrodes can be measured using a potentiostatic test at different voltages (Fig. 5b). From the fitted slopes of the voltage–current plots, the conductivity of the 40 mg cm^{−2} iodine-loaded LICA/AC electrode is found to be 0.13 S cm^{−1}, which is 1.7-folds increase compared with that of the AC electrode with the same loading. This demonstrates the superiority of the novel electrode using ionic liquid conductive agent under high active substance loading. In order to investigate the reaction kinetics in the case of thick electrodes, galvanostatic intermittent titration technique (GITT) tests are performed on batteries with thick electrodes. The transient charge/discharge curves are presented in Fig. 5c. It can be observed that under identical conditions, the coulombic efficiencies and discharge durations exhibit notable differences. The shuttle effect in the vicinity of the AC electrode is exacerbated by the imposition of a specific resting time during the test, whereas LICA/AC continues to demonstrate a high coulombic efficiency and effective suppression of the shuttle effect. The ion diffusion coefficient (D) can be calculated using the equation derived from Fick's second law (Fig. 5d).^{23,59} Calculations at different voltages demonstrate that the ion diffusion coefficient undergoes changes during the processes of charging and discharging (Fig. 5e and f). LICA/AC cathode reveals a 31-fold increase in the ion diffusion coefficient (1.82×10^{-6} cm² s^{−1} vs. 5.89×10^{-8} cm² s^{−1}) when charging cells to the charging plateau. This discrepancy can be attributed to the formation of solid iodine in the AC electrode at this stage, whereas the iodine species formed in the LICA/AC electrode will combine with the ionic liquids and predominantly exist in the form of active polyiodide ions, thus exhibiting a higher ionic mobility, which is conducive to the enhancement of ion transport during the charging process. This phenomenon is thought to be due to the Grotthuss-type homogeneous exchange mechanism.^{60,61}

To elucidate the mechanism of iodine transport in a liquid network, the process is simulated using *ab initio* molecular dynamics (AIMD).⁶² It has been demonstrated that the introduction of iodine into ionic liquids results in a notable enhancement in conductivity.⁶³ Here we take I_3^- as an example to illustrate the bond exchange process, which results in the fast movement of ions. Fig. 5g and Fig. S23 illustrate the conjugation/dissociation process of $I_3^- + I^- \rightleftharpoons I_4^{2-}$ reaction observed in the AIMD simulation. Similar to the Grotthuss transport mechanism of protons, it does not necessitate the involvement of self-diffusion or carriers in the transport process; rather, it relies on the hopping process of monatomic ions through ionization and deionization reactions. Under this circumstance, the exchange of triiodide and iodide is facilitated by the electrostatic stabilization of the surrounding ionic environment for the immediate I_4^{2-} species. An I_4^{2-} complex is formed from the iodide anion and its neighboring triiodide, which can be considered as an intermediate state with higher energy. Subsequently, the other iodide can be released from the other side. At this time, the energy of the system decreases and reaches a stable state. Each iodide atom is invariably in close coordination with an imidazole cation at the relative site of ion

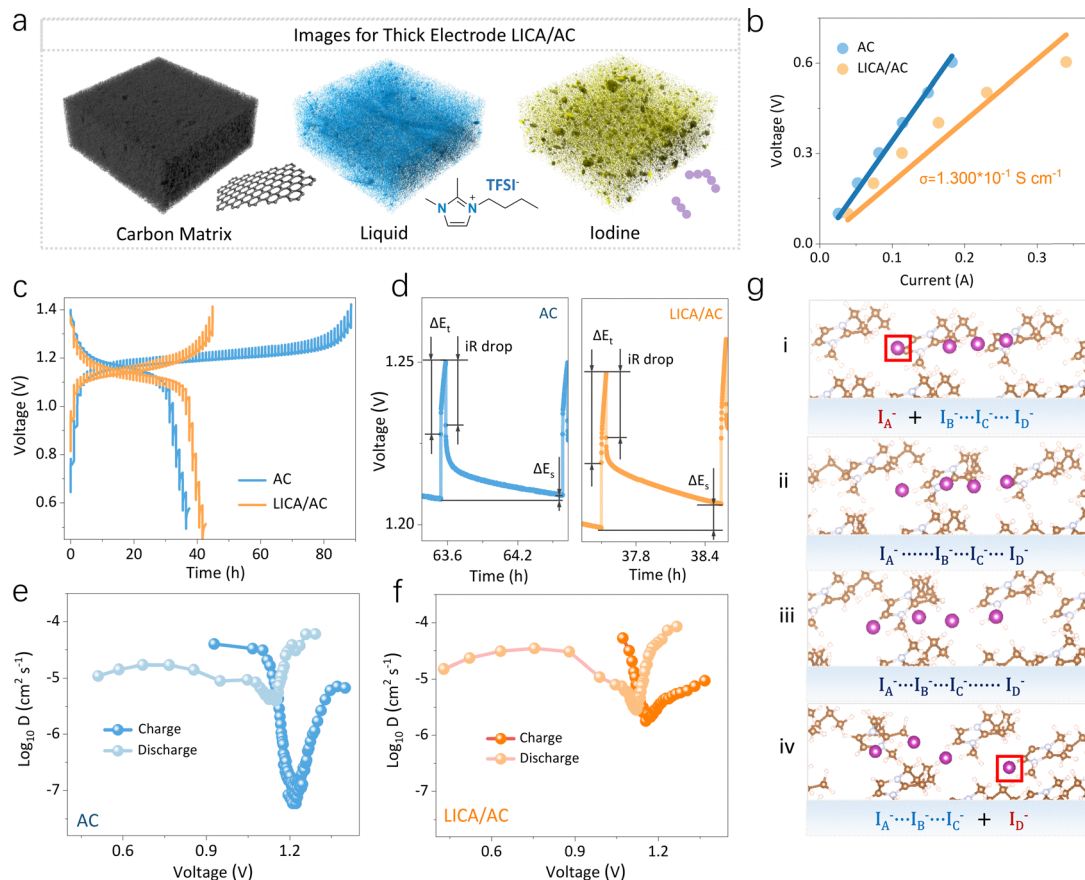


Fig. 5 (a) X-ray computer tomographic (XCT) 3D structure images of LICA/AC electrode. (b) The electrical conductivity of conditional electrodes. (c) GITT charge-discharge curves of different cells. (d) Demonstration of a single titration during a GITT measurement. (e) Calculated diffusion coefficient at different voltages during discharge of AC cell, and (f) LICA/AC cell. (g) Grotthuss transport mechanism of iodine in the reactive liquid network, showing the separated $\text{I}_3^- + \text{I}^-$ anion, the formation of the I_4^{2-} intermediate, and the release of the I^- anion on the other side and finally completing the ion transport.

environment. This result provides further evidence at the molecular level of the rapid transport of ions and high conductivity of polyhalogenated ionic liquids.

Performance of high-loading Zn-I₂ batteries

In order to fulfil the requisite high energy density for application, we further investigate the cycling performance of high-loading Zn-I₂ batteries. The initial stage of the investigation entails the exploration of varying ratios of activated carbon and iodine in the cathode. Firstly, highly iodine loaded (35 mg cm^{-2}) thick electrodes are prepared when $\text{AC/I}_2 = 0.5$, as shown in Fig. S24. At this juncture, the cell with LICA is capable of attaining a high initial areal capacity of 7.17 mAh cm^{-2} as well as a superior coulombic efficiency for nearly 50 cycles, whereas the cell utilizing the AC electrode shows an areal capacity of 6 mAh cm^{-2} and a pronounced shuttle effect after 10 cycles, thereby precluding further cycling. Fig. S25 illustrates the capacity-voltage curves of LICA/AC cells with an identical iodine loading of 40 mg cm^{-2} and different AC/I_2 ratios. It has been demonstrated that as the quantity of AC in the cathode is increased, the capacity exhibits a gradual enhancement, while the slope of the discharge curve below 1.1 V declines gradually. This phenomenon can be

attributed to the inherent capacity of the carbon itself, which manifests as the additional sloping plateaus in the discharge curve. Furthermore, porous carbon plays a crucial role in ensuring the conductivity and stability of the iodine species in the cathode. Conversely, insufficient AC can impede the full realization of iodine capacity. However, an increase in porous carbon content also results in an increase in electrode thickness. The combined capacity and electrode thickness yield the volumetric capacity. As illustrated in Fig. S26, the highest volumetric capacity can be achieved when the ratio of AC/I_2 is appropriate ($\text{AC/I}_2 = 0.66667$). Considering alongside the cycling stability shown in Fig. S27, cells with this ratio can exhibit better capacity retention while maintaining higher capacity. Consequently, this ratio is maintained for subsequent tests.

Fig. 6a illustrates that cells with a LICA/AC cathode are capable of performing charge/discharge cycles at high iodine loadings of $10, 20, 40$, and 60 mg cm^{-2} , achieving high areal capacities of $1.99, 4.02, 8.02$, and $11.55 \text{ mAh cm}^{-2}$. As the loading increases, the electrode thickness also increases significantly, resulting in a reduction in the discharge capacity of the cell due to a slowing down of the electrochemical reaction rate and a lack of utilization of the active material. However, as the

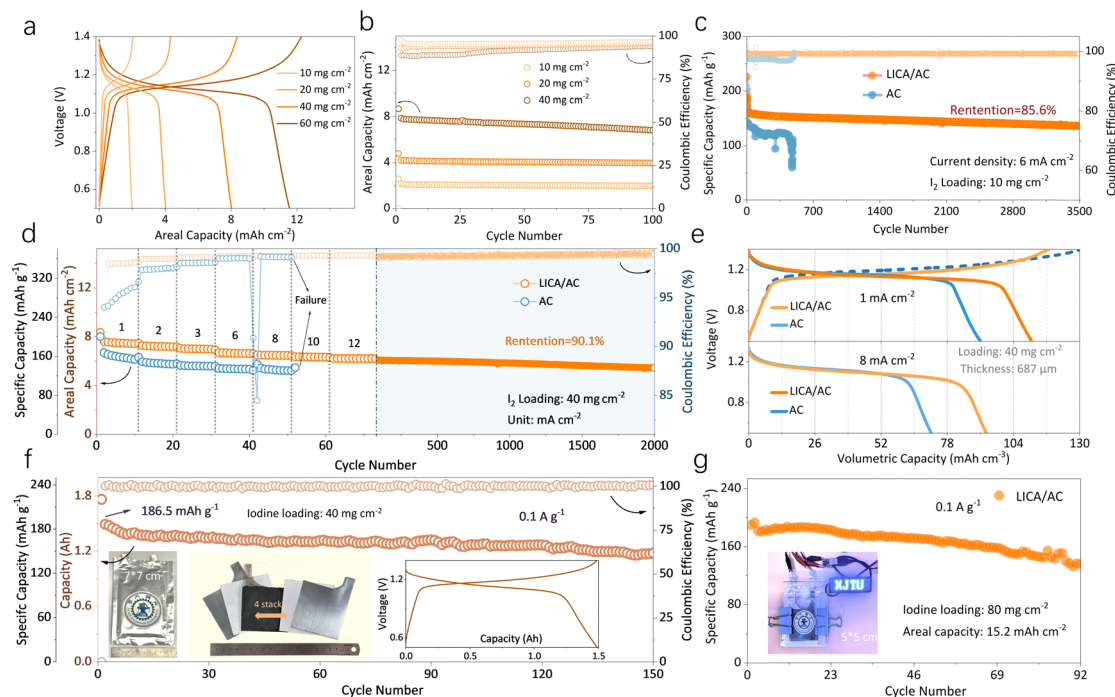


Fig. 6 (a) Galvanostatic charge–discharge (GCD) curves of LICA/AC cells with different iodine loading. (b) Cycling stability curves of LICA/AC cells with different loading. (c) Cycling performance of cells with 10 mg cm^{-2} iodine loading at 6 mA cm^{-2} . (d) Rate performance and long cycling performance of the batteries with different cathodes with 40 mg cm^{-2} iodine loading. (e) Volumetric capacity and Coulombic efficiency of AC and LICA/AC cells at different current density. (f) Long-term cycling performance of a multi-layer Zn–I₂ pouch cell of LICA/AC cathode. The inset shows the optical pictures of the pouch cell. (g) Cycling performance of Zn–I₂ pouch cell of LICA/AC cathode with ultra-high iodine loading. The inset shows the optical picture of the pouch cell acting as an energy source to power LED.

loading is increased, the specific capacity of cells using liquid conductive agent remains largely unaffected and is retained between 190 and 200 mAh g^{-1} , in accordance with the findings presented in Fig. 2g. In the case of the Zn–I₂ battery, the application of small currents results in an increase in the cycling time and a concomitant worsening of the shuttle effect. Fig. 6b and Fig. S28 show the cycling stability curves of Zn–I₂ cell with the LICA/AC cathode at 1 mA cm^{-2} with varying iodine loadings, all of which are capable of ensuring at least 100 cycles with minimal attenuation. Fig. 6c illustrates the results of a high-current long-cycle test conducted with a loading of 10 mg cm^{-2} . In contrast to Zn–I₂ cells with an AC cathode, which exhibits a rapid decay in capacity after approximately 500 cycles, those of LICA/AC can maintain a capacity retention of over 85% for 3500 cycles and the coulombic efficiency reaches approximately 100%.

We explore the cycling performance of Zn–I₂ cell with high iodine loading. Fig. 6d, e and Fig. S29 present the comparison of the rate performance of Zn–I₂ cells at 40 mg cm^{-2} utilizing two distinct cathodes. It is noteworthy that the Zn–I₂ cells with LICA/AC possess enhanced capacity at all current densities and exemplary rate performance in comparison to those with AC. The initial capacity of the Zn–I₂ cells with AC at 1 mA cm^{-2} is 6.26 mAh cm^{-2} and the initial coulombic efficiency is only 69.83%. In contrast, the initial coulombic efficiency of LICA/AC cells is 94.33% with a far higher areal capacity, which provides further evidence that the introduction of a liquid conductive

agent has a strong inhibitory effect on the shuttle effect even at small current density. Cells assemble AC cathode exhibit overcharging at 8 mA cm^{-2} , which can be attributed to the shuttle effect, while at this current density the cell of LICA/AC still achieves a high areal capacity of 6.41 mAh cm^{-2} with high coulombic efficiency. It should be noted that following the completion of rate performance testing, the current density is stabilized at 12 mA cm^{-2} for long-term cycling testing ($\approx 7 \text{ min}$ for charging), where the cells with LICA/AC continue to demonstrate an impressive areal capacity of 6.30 mAh cm^{-2} and stable cycling for 2000 cycles, with a capacity retention of 90.1%. Compared to other representative work in the literature, the LICA/AC electrode shows ultra-high iodine loading and high iodine content in I₂/AC, and the battery system shows competitive electrochemical performance especially in terms of areal capacity (Fig. S30 and Table S1). With regard to volumetric capacity, zinc–iodine battery with LICA demonstrates an increase to 122% ($111.2 \text{ mAh cm}^{-3}$ vs. 91.0 mAh cm^{-3}) and 130% (93.3 mAh cm^{-3} vs. 71.5 mAh cm^{-3}) compared with conventional porous carbon cathode at 1 mA cm^{-2} and 8 mA cm^{-2} , respectively (Fig. 6e). This demonstrates that LICA can improve the electrical conductivity and mediate confined transport of iodine in thick electrodes without adding extra electrode thickness and sacrificing volumetric capacity.

To further exemplify the practical implementation of innovative cathode for grid-scale energy storage, we devise a $7 \times 7 \text{ cm}$ Zn–I₂ pouch cell at the ampere-hour-level comprising a

multi-layer stacked structure (Fig. 6f). This pouch cell demonstrates a high discharge capacity of approximately 1.5 Ah (7.65 mAh cm^{-2}) and a stable cycling over 150 cycles, providing a reliable basis for the practical application of Zn-I₂ batteries. Subsequently, we produce a $5 \times 5 \text{ cm}$ pouch cell using LICA/AC electrode with double-loaded up to 80 mg cm^{-2} , which achieves an areal capacity of $15.20 \text{ mAh cm}^{-2}$ and a specific capacity of nearly 190 mAh g^{-1} (Fig. 6g and Fig. S31). The overall capacity of the pouch cell reaches as high as 380 mAh, nearly 400 mWh, with a successful cycling rate of nearly 100 cycles. This exhibits an excellent performance compared with other reported work, as shown in Fig. S32 and Table S2.

Conclusions

In this contribution, we used IL as a liquid ionic conductive agent (LICA) in an iodine cathode, exploiting the robust adsorption and high conductivity of IL on iodine species, with the objective of addressing the issue of thick electrodes in Zn-I₂ batteries under high loading conditions. The introduction of LICA results in inhibition of the dissolution and shuttling of polyiodides, and increase in the ionic conductivity of the cathode. To demonstrate the efficacy of this approach, ultra-high loaded Zn-I₂ batteries are assembled. A zinc-iodine battery with an ionic liquid conductive agent reaches a 7-fold increase in cycle life (3500 cycles) at 10 mg cm^{-2} iodine loading, and 40-fold increase (2000 cycles) at 40 mg cm^{-2} , with an initial areal capacity of 6.30 mAh cm^{-2} and a capacity decay of 0.00495% per cycle ($\approx 7 \text{ min}$ for charging). An ultra-high loading of 60 mg cm^{-2} with an areal capacity of $11.55 \text{ mAh cm}^{-2}$ can also be achieved. Optimising the cathode allows for the stable cycling of a single-layer pouch cell with 80 mg cm^{-2} double-loaded iodine, as well as a multi-layer stacked pouch cell at the ampere-hour level. The present study offers novel design concepts for the fabrication of highly loaded thick electrodes for Zn-I₂ batteries and provides evidence supporting the viability of Zn-I₂ batteries in real-world applications.

Author contributions

All authors have given the permission for the final version of this manuscript. Conceptualization and visualization: L. Zhao, H. Zhao; methodology and investigation: L. Zhao, D. Yin, Y. Zhang, X. Cui, J. Feng, N. Gao. Formal analysis: B. Li, S. Wang. Validation and writing – original draft: L. Zhao; writing – review, and editing: L. Zhao, S. Ding, H. Zhao. Funding acquisition: X. Liu, S. Ding. Project administration and supervision: S. Ding.

Conflicts of interest

There are no conflicts of interest to declare.

Data availability

The data supporting this article have been included as part of the SI. The Supplementary Information file mainly contains descriptions of various characterisation methods for materials, characterisation results, battery assembly methods, and demonstrations of additional battery performance. See DOI: <https://doi.org/10.1039/d5ee02941d>.

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